

5 Resistivity

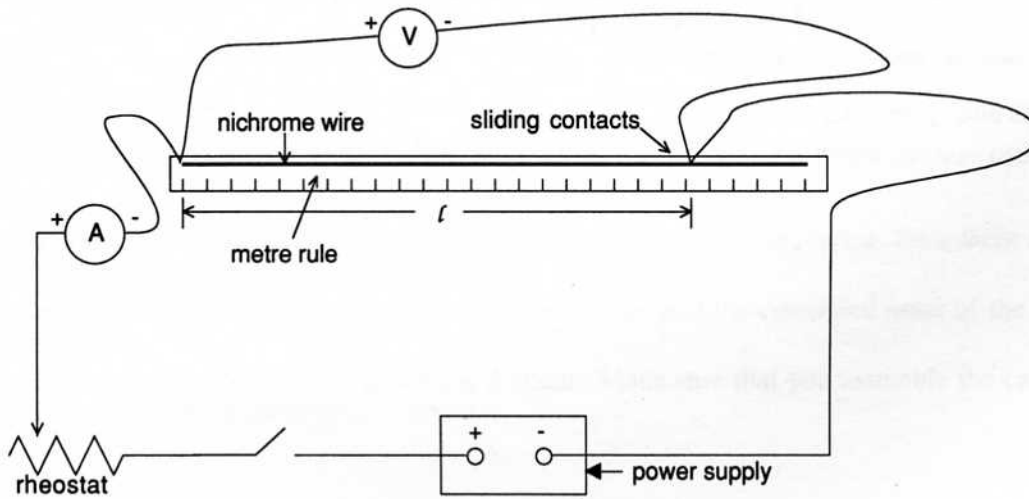
The resistivity of a conductor of length l metres, cross-sectional area A square metres, and resistance R ohms is:

$$\rho = \frac{RA}{l}$$

Resistivity is a characteristic of a conductor.

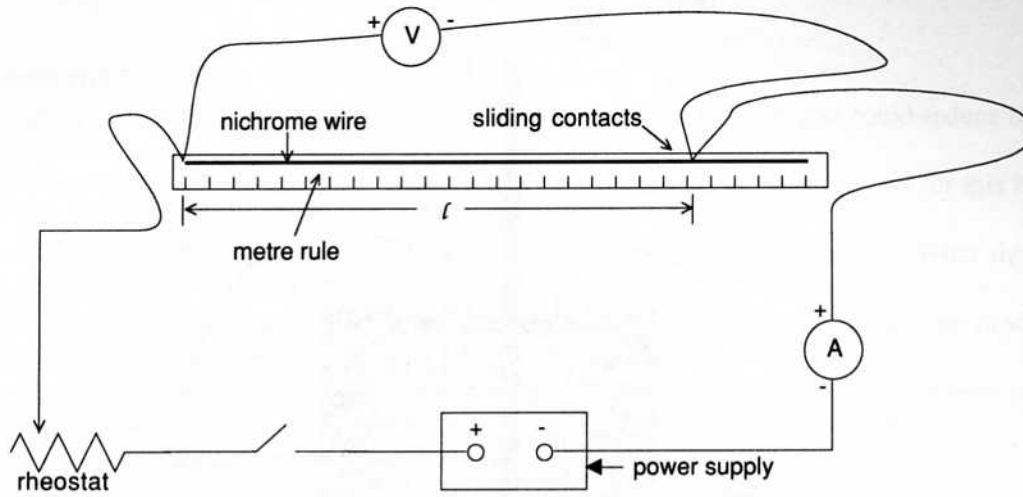
You can determine the resistance of the section of the wire in the circuit diagram below using:

$$R = \rho \frac{l}{A}$$



In this experiment you will calculate the resistivity of a piece of thin wire.

- 2 Explain if the position of the ammeter in the following diagram is preferable to the position of the ammeter in the original circuit.



- 3 Which of the various measurements has the greatest potential for producing error in your final result? Suggest a more precise method for making this measurement.
- 4 If you use thicker wire, how would you expect the following values to change:
- a the ratio of the resistance of the wire to its length;
 - b the wire's resistivity.